28 (2026) 100663

Contents lists available at ScienceDirect

The Journal of Climate Change and Health

journal homepage: www.elsevier.com/locate/joclim

Research article

# Climate change and youth in South Korea: A gendered analysis of psychosocial and behavioral profile

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A R T I C L E I N F O

*Keywords:*
Climate anxiety
Climate emotions
Climate action
Climate education
Young people
Survey

A B S T R A C T

*Introduction:* Understanding how youth perceive and respond to climate change is essential for designing effective education and engagement strategies. This study assessed the psychosocial and behavioral profiles related to climate change among Korean youth.
*Materials and Methods:* Data from the Youth-led Carbon Neutrality Promotion Survey conducted by the National Youth Policy Institute ($n$ = 3024; age=11–18 years) were used. Descriptive and logistic regression analyses were conducted, adjusting for school grade and type (co-ed vs gender-segregated).
*Results:* Most Korean youth believed that climate change is currently happening (83%) and caused by human activity (95%). Boys showed higher technical knowledge (e.g., familiarity with Net-Zero terminology) but were less likely than girls to believe climate change is happening, feel climate anxiety, or take personal pro-environmental actions. Conversely, boys were more optimistic and viewed institutional responses as effective. These patterns suggest girls engage more emotionally and behaviorally, while boys focus on policy-level solutions, which may contribute to lower psychosocial response and action. Youth placed importance on both systemic (e.g., government policies, climate education) and individual-level strategies (individual-level information, youth engagement) as potential solutions, but less importance was placed on incentive-based or punitive measures. Using public transportation emerged as the most frequently adopted climate action (48%), far surpassing other behaviors.
*Discussion:* Findings suggest gendered pathways in climate engagement, primarily girls' affective-behavioral responses and boys' structural-policy optimism.
*Conclusion:* These findings highlight the importance of tailored climate education and action programs that consider gendered patterns in psychosocial and behavioral profile to more effectively support youth as agents of climate action.

## 1. Introduction

Climate change is one of the most pressing global challenges of our time, with significant implications for current and future generations [1–3]. Youth, as the generation of leaders, policymakers, and citizens, play a key role in shaping climate action [4]. Given that climate change is a global issue that requires coordinated global action, there is a pressing need to foster international youth advocacy and collaboration. For instance, in the European Union, youth climate action is supported through policies such as the Green Deal and youth strategies, with an emphasis on meaningful participation in climate governance, capacity-building for policy engagement, and support for youth-led initiatives [5]. In South Korea (Korea thereafter), recent policy developments including the enactment of the Framework Act on Carbon

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https://doi.org/10.1016/j.joclim.2026.100663
Received 9 October 2025; Accepted 4 February 2026
Available online 5 March 2026

Neutrality and the revision of the Framework Act on Education indicate a growing recognition of youth engagement in climate action [6]. These efforts aim to ensure youth rights and well-being within carbon neutrality strategies, expand school-based environmental education, foster collaboration across relevant government agencies, and strengthen the integration of climate issues with youth concerns such as welfare, safety, and future employment.

Previous research has indicated that psychosocial and behavioral responses to climate change such as perceptions, knowledge, attitudes, and actions, collectively referred to as “social climate” in this study, are critical factors that determine the effectiveness of future climate policies and initiatives [7,8]. Despite the growing recognition of youth engagement in climate discourse, research on the social climate of climate change among youth remains limited. Most previous studies on the social climate of climate change have focused primarily on the young adult or adult populations [4,9,10], and, when available, evidence is limited by non-generalizable data [4]. Nevertheless, there is a growing interest in this topic globally. In a 2024 cross-sectional descriptive survey involving 15,793 United States citizens aged 16–25 years [11], 85 % of respondents indicated that they are worried about climate change and its impact on people and the planet, with 58% being extremely worried. Furthermore, almost half of the respondents indicated that climate change is negatively impacting their mental health, highlighting the need for more research on how youth perceive and engage with climate change.

The Intergovernmental Panel on Climate Change (IPCC) has identified Northeast Asia, including Korea, as a climate-vulnerable region due to its high population density, industrialization, and susceptibility to extreme climate events [12]. Indeed, Korea faces unique environmental challenges, including air pollution, typhoons, heavy rains, droughts, and abnormal temperatures [13]. With aggregating climate change impacts, understanding how Korean youth perceive and engage with climate change is essential for developing targeted educational interventions and policies that empower youth to take meaningful climate action. Recent research examined the growing climate activism movement among Korean youth by highlighting their civic engagement, participation in policy processes, and mobilization through social networks [14–17]. These studies provided insights into the sociopolitical dimensions of youth climate action and the structural barriers and facilitators shaping their participation. However, to date, no studies have comprehensively explored the psychosocial and behavioral profile of climate change among Korean youth. At the same time, emerging climate and social science research consistently shows gendered differences in cognitive, emotional, and behavioral responses to climate change [18–20], which may highlight its importance on informing inclusive and effective public strategies in a rapidly changing environmental context.

In this paper, perception and engagement in climate change are operationalized following the framework proposed by Van Valkengoed and colleagues [21], which conceptualizes individual perceptions along three interrelated dimensions: 1) cognitive dimensions of risk (e.g., knowledge about climate change); 2) experiential processes (e.g., affect, personal experience); and 3) sociocultural influences (e.g., value orientations, social norms). The model lays the groundwork for more nuanced, evidence-based approaches to understanding climate engagement. Specifically, belief in climate change, attribution of its causes, as well as broader perceptual and affective dimensions, such as climate awareness, climate anxiety, and affect, can be used to understand how young people internalize, emotionally respond to, and act upon their understanding of climate change [21,22]. Measures such as perceived sufficiency of school-based education, national response effectiveness, and optimism about solving climate change also reflect young people’s appraisals of societal and institutional efficacy, which may impact their sociocultural influences. Building on previous theorization [21,22], better understanding of how cognitive (e.g., belief, attribution of causes, knowledge), experiential (e.g., affect, anxiety), and sociocultural (e.g., value, action, perceptions of institutional effectiveness) dimensions occur among young people is important. Therefore, this study examined the psychosocial and behavioral profile of climate change among 3, 024 Korean youth aged 10–18 years. This study also examined whether these factors differ by gender, given that potential gendered patterns are rarely understood among Korean youth. Adopting a gendered lens in better understanding perception and engagement with climate change is important in informing the design of equitable and targeted approaches to youth engagement in climate action.

## 2. Materials and methods

### 2.1. Data

The Youth-led Carbon Neutrality Promotion Survey conducted by the National Youth Policy Institute (NYPI) [6] was used. The purpose of this cross-sectional survey was to identify strategies for ensuring the rights and promoting the participation of youth in the transition to a carbon-neutral society. A self-administered questionnaire was used to collect data over a four-month period (May–August 2022) among students aged 10–18 years. Student consent was obtained prior to survey administration, and trained research staff conducted the data collection on site. A total of 3, 024 responses were collected and considered for analysis. The study protocol was approved by the National Youth Policy Institute’s ethics board (Approval No 202,204-HR-Unique-004) [6].

### 2.2. Measures

Key variables were selected based on previous work theorizing climate perceptions [4,21,22]. Development process and validation of the survey items are detailed elsewhere [6], and in the Supplementary File.

#### 2.2.1. Key variables

Key variables included measures of climate belief, causal attribution, knowledge, awareness, anxiety, and self-efficacy, as well as perceptions of school-based climate education, national response effectiveness, optimism about solving climate change, willingness for behavioral modification, prioritized climate solutions, and prior climate action. Detailed information about measurement variables, definition/conceptualization, items, response options, categorization for analysis, and psychometric information (when reported) are available in Table 1.

#### 2.2.2. Covariates

School grade (grade 5 to grade 12) and type (co-ed, girls only, boys only) were included as covariates.

### 2.3. Statistical analysis

Descriptive analyses were conducted on the study variables. Frequencies and proportions were calculated for categorical variables, while means and standard deviations were reported for continuous variables. To examine the association between gender and climate-related variables, logistic regression analyses were performed, adjusting for potential confounders, including school grade and school type (e.g., co-ed vs. gender-segregated). Adjusted odds ratios (AORs) and 95 % confidence intervals (CIs) were estimated for each of the climate-related variable as the outcome. Statistical significance was determined using a two-tailed p-value threshold of $<0.05$. All analyses were conducted using SPSS version 29.0 (IBM Corp., Armonk, NY, USA), employing the Complex Samples function to account for the stratified, multi-stage design and probability sampling that account for area of residence and school level. Google Colaboratory, a free, cloud-based coding platform that enables writing and executing Python code in a web-based Jupyter Notebook environment, was used to create figures presented in this work.

**Table 1**
Measurement of key study variables.

| Variable | Definition/ Conceptualization | Measurement Items | Response Options | Gender-Based Categorization/Coding | Reliability (Cronbach's α) |
|---|---|---|---|---|---|
| Climate Belief | Belief regarding whether climate change is currently happening. | "Do you believe climate change is happening?" | (1) Currently happening; (2) Not happening but will in 5 years; (3) Not happening but will in 10 years; (4) Not happening and never will; (5) I don't know | Categorized as "Currently happening" vs. "Currently not happening"; "I don't know" = missing | NA |
| Causal Attribution of Climate Change | Beliefs about the primary cause of climate change. | "What do you think causes climate change?" | (1) Human activity; (2) Natural phenomenon | Dichotomous: Human activity vs. Natural phenomenon | NA |
| Climate Knowledge | Factual understanding of causes, impacts, and solutions of climate change. | Familiarity with 7 terms (e.g., Net-Zero, Paris Agreement, climate inequality). | (1) Heard of it; (2) Heard but don't know meaning; (3) Heard and understand well | Continuous or summary score; higher = greater knowledge | NA |
| Climate Awareness | Awareness, interest, and perceived seriousness of climate change and actions. | 5 items (e.g., interest in climate change, perceived seriousness, knowledge of adaptation). | 5-point Likert scale (Strongly disagree–Strongly agree) | Summary score averaged; dichotomized: <4.0 = low awareness; ≥4.0 = high awareness | 0.74 |
| Climate Anxiety | Emotional distress, worry, or fear related to climate change. | 7 items reduced to 6 after reliability testing (e.g., anxious, helpless, angry, guilty). | 5-point Likert scale (Strongly disagree–Strongly agree) | Summary score averaged; dichotomized: <4.0 = low anxiety; ≥4.0 = high anxiety | 0.74 (after removing 1 item) |
| Climate Self-Efficacy | Confidence in one's ability to take climate-related actions. | 20 items assessing knowledge, responsibility, problem-solving, communication, and collaboration. | 5-point Likert scale (Strongly disagree–Strongly agree) | Summary score averaged; dichotomized: <4.0 = low efficacy; ≥4.0 = high efficacy | 0.93 |
| Sufficiency of School-based Climate Education | Perceived adequacy of school climate education. | "Climate change education is adequately provided at school." | (1) Not at all; (2) No; (3) Neutral; (4) Yes; (5) Very much so; (6) I don't know | Categorized: No/Neutral vs. Yes/Very much so; "I don't know" = missing | NA |
| National Climate Response Effectiveness | Perceived adequacy of national response to climate change. | "Our country is responding well to climate change." | (1) Not at all; (2) No; (3) Neutral; (4) Yes; (5) Very much so; (6) I don't know | Categorized: No/Neutral vs. Yes/Very much so; "I don't know" = missing | NA |
| Optimism about Solving Climate Change | Belief in humanity's ability to solve climate change. | "Ultimately, humanity will solve the problem of climate change." | (1) Not at all; (2) No; (3) Neutral; (4) Yes; (5) Very much so; (6) I don't know | Categorized: No/Neutral vs. Yes/Very much so; "I don't know" = missing | NA |
| Willingness for Behavioral Modification | Willingness to modify daily behaviors to help solve climate change. | "Select the option closest to your current lifestyle habits related to solving climate change problems." | 5-point Likert scale (Not willing at all-Very willing) | Categorized: Not willing (<4) vs. Willing (≥4) | NA |
| Prioritized Climate Solution | Participants' perceived most important solutions to climate change. | "Select and rank two most important factors for solving climate change." | Nine options (e.g., information, education, youth participation, policy, corporate action). | First-ranked responses analyzed | NA |
| Climate Action | Participation in individual, collective, or systemic climate actions. | 13 activities (e.g., campaigns, energy reduction, youth programs, plant-based diet). | Multiple choice | Grouped into: Activism / Pro-environmental behaviors / No participation | NA |
| Covariates | Demographic and school characteristics. | Grade (5–12), school type (co-ed, girls-only, boys-only). | NA | NA | NA |

NA: Not applicable.

## 3. Results

Of 3, 024 who responded to the survey, 2, 586 youth provided complete data on the variables used in this study. No statistically significant differences were observed for area of residence, school level and grade or school type between the participants included and excluded. Table 2 describes participant characteristics and effects on the study variables. In terms of climate belief, a strong majority (83.3 %) of youth believed climate change is currently happening. Boys were slightly more likely than girls to have an expectation of delayed climate change effects (higher in the 5- and 10-year categories). A near-unanimous agreement across all groups (94.7 %) in the belief that human activity is the primary cause of climate change was found. Belief in natural causes was minimal (5.3 %) and consistent across gender.

Climate knowledge varied widely across specific topics (Fig. 1). While the vast majority of youth (95.0 %) reported knowing about global warming, knowledge of terms like net-zero (27.7 %) and climate inequality (24.8 %) was considerably lower. Awareness of sustainable development (51.7 %) and climate refugees (51.2 %) was moderate, with little difference between girls and boys. When asked to identify the most important solution to climate change (Fig. 2), youth most frequently selected government policies (22.3 %), followed by individual-level information (15.1 %) and climate education (14.5 %). Girls and boys shared similar priorities. Support for green business practices (11.5 %) and stronger penalties (11.0 %) was moderate, while fewer respondents emphasized global cooperation (8.3 %) or incentives for action (5.8 %) as top solutions. Fig. 3 illustrates youth engagement in climate action. The most commonly reported action was using public transit to and from school, with nearly half (48.0 %) of youth indicating participation. Other relatively common actions included participation in local environmental cleanup (13.3 %) and school-based carbon reduction activities (11.3 %). Less frequently reported actions included involvement in climate crisis awareness campaigns (6.0 %), reducing home energy use (5.3 %), and adopting a plant-based diet (3.3 %). Only a small proportion of youth reported participating in more organized or collective forms of action, such as engaging in out-of-school climate activities (0.4 %), proposing climate policies (0.2 %), joining youth climate clubs (0.1 %), or participating in youth exchange activities (0.2 %). Despite these varied actions, 7.5 % of youth reported no participation or experience with climate action, with boys more likely than girls to fall into this group (9.6 % vs. 5.7 %). Gender-stratified percentages are shown in Figs. 1 and 3, but are not discussed in detail here, as no statistical comparisons between genders were conducted.

Table 3 shows gender differences in the psychosocial and behavioral

**Table 2**
Participant characteristics ($n = 2586$).

| | All | Girls (52.9 %; $n$ = 1369) | Boys (47.1 %; $n$ = 1217) |
|---|---|---|---|
| **Area of residence** ( %)† | | | |
| Seoul | 17.1 | 17.1 | 17.1 |
| Gyeonggi/Incheon | 32.0 | 31.8 | 32.3 |
| Gangwon/Chungcheong | 13.2 | 13.5 | 13.0 |
| Kyongsang | 24.6 | 24.7 | 24.5 |
| Joella/Jeju | 13.0 | 12.9 | 13.2 |
| **School grade** ( %)† | | | |
| Grade 5 | 12.3 | 12.5 | 12.2 |
| Grade 6 | 12.8 | 12.9 | 12.7 |
| Grade 7 | 12.4 | 12.5 | 12.4 |
| Grade 8 | 12.9 | 12.9 | 12.9 |
| Grade 9 | 12.4 | 12.9 | 11.9 |
| Grade 10 | 11.2 | 13.5 | 8.6 |
| Grade 11 | 13.5 | 15.6 | 11.3 |
| Grade 12 | 12.4 | 7.2 | 18.1 |
| **School type** ( %)† | | | |
| Co-Ed | 77.9 | 80.1 | 75.5 |
| Girls only | 10.6 | 19.9 | - |
| Boys only | 11.5 | - | 25.5 |
| **Cognitive Factors** | | | |
| *Climate belief* ( %, 95 % CI) | | | |
| Currently happening | 83.3 (77.5–87.8) | 85.3 (77.9–90.5) | 81.0 (75.8–85.3) |
| Not currently happening but expected in 5 years | 7.2 (4.8–10.5) | 6.4 (3.8–10.8) | 8.0 (5.511.5) |
| Not currently happening but expected in 10 years | 3.6 (3.1–4.7) | 2.7 (1.5–4.9) | 5.0 (4.1–6.2) |
| Not currently happening and will not occur | 0.4 (0.2–0.8) | 0.1 (0.0–1.5) | 0.7 (0.4–1.1) |
| Do not know | 5.4 (3.7–7.7) | 5.4 (3.7–7.9) | 5.3 (3.6–7.8) |
| *Causal attribution* ( %, 95 % CI) | | | |
| Human activity | 94.7 (92.8–96.1) | 94.7 (91.2–96.8) | 94.7 (93.3–95.9) |
| Natural phenomenon | 5.3 (3.9–7.2) | 5.3 (3.2–8.8) | 5.3 (4.1–6.7) |
| *Climate knowledge (know)* ( %, 95 % CI) | | | |
| Net-Zero | 27.7 (19.9–37.2) | 23.8 (16.4–33.2) | 32.1 (23.5–42.2) |
| Paris agreement | 31.8 (20.2–46.2) | 30.9 (18.8–46.4) | 32.8 (21.5–46.6) |
| Greenhouse gas emission reduction | 48.5 (36.8–60.3) | 46.0 (35.2–57.0) | 51.3 (38.0–64.3) |
| Sustainable development | 51.7 (36.9–66.2) | 50.3 (35.4–65.1) | 53.3 (38.0–67.9) |
| Global warming | 95.0 (93.4–95.2) | 95.2 (93.4–96.5) | 94.9 (93.5–96.0) |
| Climate refugee | 51.2 (42.6–59.7) | 50.9 (41.2–60.7) | 51.4 (43.9–58.9) |
| Climate inequality | 24.8 (19.9–30.5) | 25.0 (19.5–31.4) | 24.6 (19.6–30.3) |
| **Experiential Factors** | | | |
| *Climate awareness* (1–5) (Mean ± SE) | 3.28 ± 0.04 | 3.30 ± 0.05 | 3.25 ± 0.04 |
| *Climate anxiety* (1–5) (Mean ± SE) | 3.00 ± 0.02 | 3.08 ± 0.03 | 2.91 ± 0.02 |
| *Environmental competency* (1–5) (Mean ± SE) | 3.00 ± 0.03 | 3.05 ± 0.03 | 2.94 ± 0.03 |
| **Sociocultural Factors** | | | |
| *Sufficiency of school-based climate education* (1–5) (Mean ± SE) | 3.28 ± 0.07 | 3.27 ± 0.08 | 3.29 ± 0.06 |
| *National climate response effectiveness* (1–5) (Mean ± SE) | 2.65 ± 0.05 | 2.56 ± 0.06 | 2.75 ± 0.05 |
| *Optimism about solving climate change* (1–5) (Mean ± SE) | 2.77 ± 0.04 | 2.67 ± 0.07 | 2.87 ± 0.04 |
| *Prioritized climate solution* ( %, 95 % CI) | | | |
| Individual level info | 15.1 (13.1–17.3) | 15.0 (12.3–18.3) | 15.1 (12.7–17.9) |
| Climate change education | 14.5 (13.0–16.2) | 13.9 (11.7–16.5) | 15.1 (13.0–17.6) |
| Youth engagement | 11.3 (8.6–14.7) | 11.6 (8.0–16.5) | 11.0 (8.9–13.4) |
| Stronger penalties | 11.0 (9.6–12.5) | 11.0 (8.9–13.5) | 10.9 (8.6–13.8) |
| Incentives for action | 5.8 (4.9–6.9) | 6.0 (4.3–8.4) | 5.6 (5.1–6.1) |
| Green business practices | 11.5 (9.3–14.1) | 11.4 (8.3–15.3) | 11.7 (9.7–13.9) |
| Global cooperation | 8.3 (6.2–11.1) | 8.1 (5.6–11.5) | 8.7 (6.6–11.3) |
| Government policies | 22.3 (19.9–24.8) | 22.8 (19.1–26.9) | 21.7 (18.4–25.4) |
| **Behavioral factors** | | | |
| *Behavioral modification* (1–5) (Mean ± SE) | 2.82 ± 0.06 | 2.90 ± 0.05 | 2.74 ± 0.07 |
| *Climate action* ( %, 95 % CI) | | | |
| School carbon reduction activities | 11.3 (8.9–14.3) | 9.8 (7.8–12.3) | 12.9 (9.3–17.7) |
| Use public transit to/from school | 48.0 (43.4–52.7) | 54.3 (48.0–60.5) | 41.0 (36.5–45.6) |
| Climate crisis awareness campaign | 6.0 (4.6–7.8) | 5.9 (3.8–9.0) | 6.2 (4.6–8.4) |
| Plant-based diet (reduce meat) | 3.3 (2.4–4.7) | 3.6 (2.2–5.8) | 3.0 (2.1–4.2) |
| Local environmental cleanup | 13.3 (11.5–15.3) | 12.1 (10.1–14.5) | 14.6 (11.9–17.9) |
| Youth climate exchange activities | 0.2 (0.1–0.8) | 0.4 (0.1–1.5) | 0.4 (0.1–1.5) |
| Propose policies to address climate change | 0.2 (0.1–0.6) | 0.2 (0.1–1.0) | 0.2 (0.0–0.8) |
| Support eco-conscious companies | 1.2 (0.8–1.8) | 1.3 (0.9–2.1) | 1.0 (0.4–2.5) |
| Reduce home energy use | 5.3 (3.5–8.1) | 4.0 (2.5–6.4) | 6.8 (4.5–10.2) |
| Reduce or sort household waste | 3.1 (2.4–4.0) | 2.5 (1.5–4.0) | 3.8 (3.0–4.8) |
| Join/form climate youth clubs | 0.1 (0.0–0.3) | 0.0 (0.0–100.0) | 0.1 (0.0–0.7) |
| Out-of-school climate activities | 0.4 (0.2–0.9) | 0.4 (0.2–1.2) | 0.3 (0.1–0.9) |
| No participation or experience | 7.5 (5.7–9.9) | 5.7 (3.6–9.0) | 9.6 (7.3–12.4) |

† Based on the total, non-complex sampling ($N = 3024$).
Data are presented as percentages with 95 % confidence intervals (95 % CI) or means with standard errors (SE).

profiles of climate change. Compared to girls (the reference group), boys showed lower odds of believing that climate change is currently happening (OR: 0.68; 95 %CI: 0.48–0.97). However, there were no statistically significant gender differences in causal attribution, as boys and girls were similarly likely to attribute climate change to human activity rather than natural phenomena. In terms of climate knowledge, boys showed higher odds of reporting knowing about the concept of "Net Zero" (OR: 1.47; 95 % CI: 1.17–1.85), while no statistically significant gender differences were observed for knowledge of the Paris Agreement, greenhouse gas emission reduction, sustainable development, global warming, climate refugees, or climate inequality. There was no gender difference in self-reported climate awareness. However, boys were less likely to report high levels of climate anxiety compared to girls (OR: 0.63; 95 % CI: 0.41–0.95). Although boys also showed lower odds of reporting high climate self-efficacy (OR: 0.74; 95 % CI: 0.53–1.05), this result did not reach statistical significance. In terms of perceived institutional responses, boys were more likely than girls to believe that national-level responses to climate change were effective (OR: 2.19; 95 % CI: 1.67–2.88). They also reported greater optimism that climate change could be addressed successfully (OR: 1.96; 95 % CI: 1.36–2.71). As for action, boys were less likely than girls to report modifying their behaviors in response to climate change (OR: 0.74; 95 % CI: 0.57–0.94).

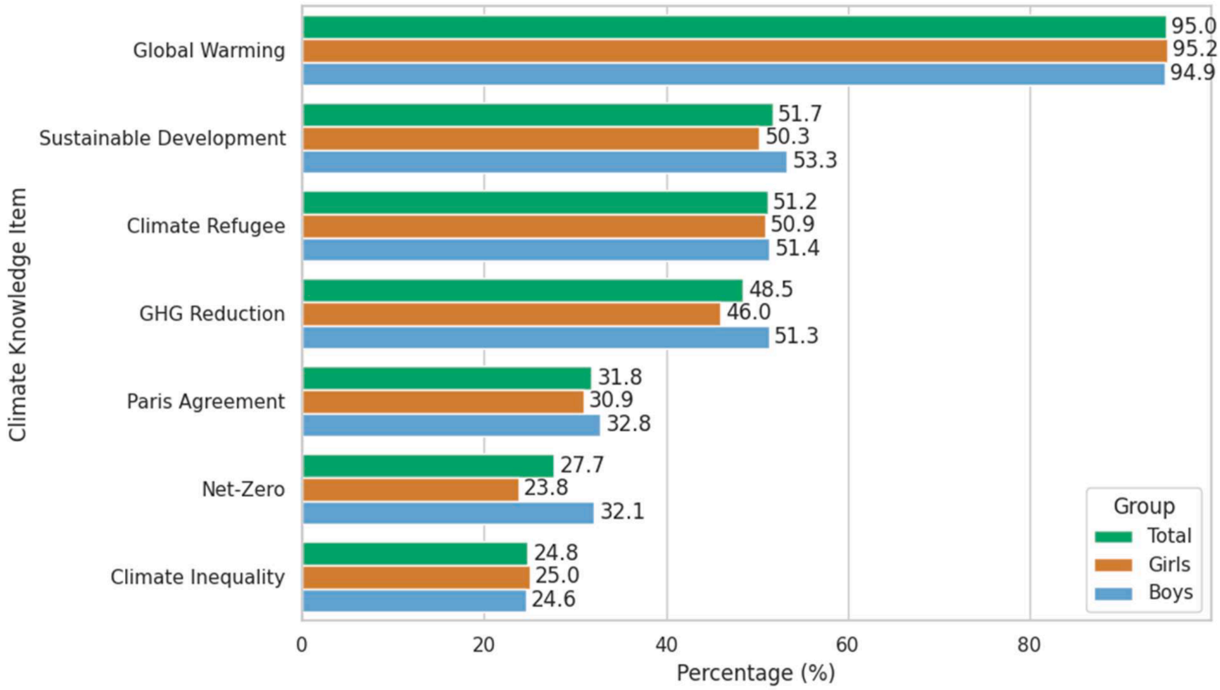


**Fig. 1.** Climate knowledge by gender.

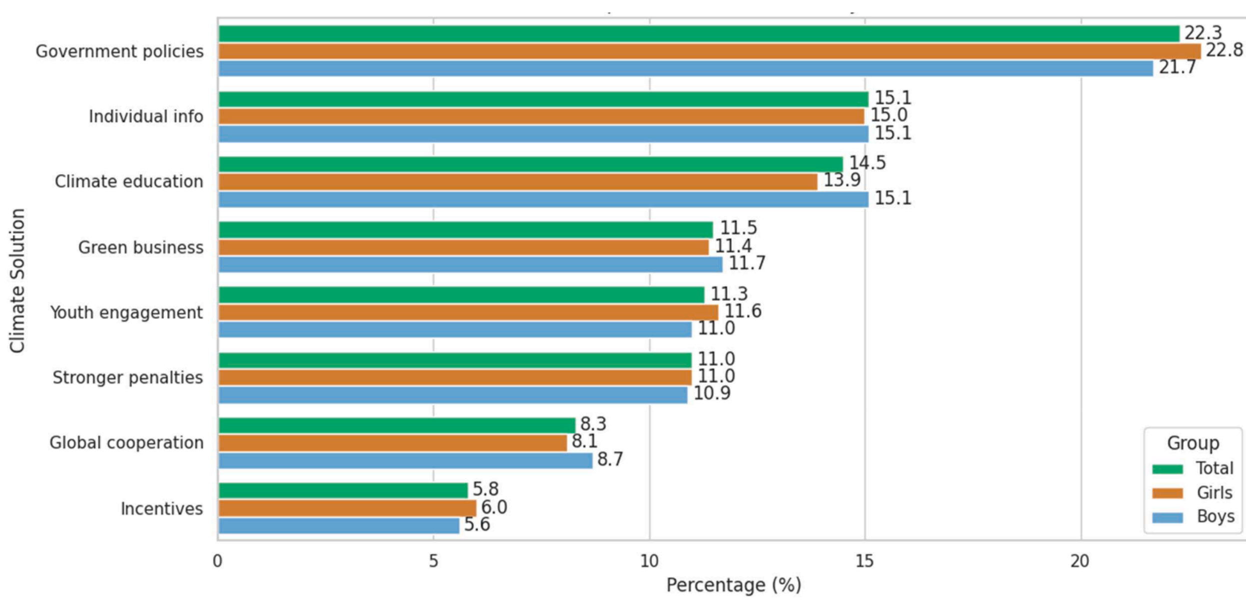


**Fig. 2.** Prioritized climate solution by gender.

However, no statistically significant gender differences were found regarding the perceived sufficiency of school-based climate education. Regarding beliefs about the most important climate solutions, there were no statistically significant gender differences in prioritizing individual-level information, climate education, youth engagement, penalties, incentives, green business practices, global cooperation, or government policies. For climate-related action, boys were less likely to report engaging in pro-environmental behaviors (OR: 0.64; 95 % CI: 0.46–0.88). Boys also showed higher odds of reporting no participation in climate-related actions (OR: 1.82; 95 % CI: 1.14–2.91).

## 4. Discussion

This study examined gendered psychosocial and behavioral profile on climate change among Korean youth. Our findings suggested that while boys report higher knowledge in specific technical areas (e.g., Net-Zero), they are less likely than girls to believe climate change is happening, to feel climate anxiety, or to engage in personal pro-environmental actions. Conversely, boys were more optimistic and more likely to perceive institutional responses as effective. This divergence suggests that girls may engage with climate change through affective and behavioral pathways, while boys show higher confidence in structural or policy-level solutions, potentially leading to lower emotional response or action. These patterns highlight the need for

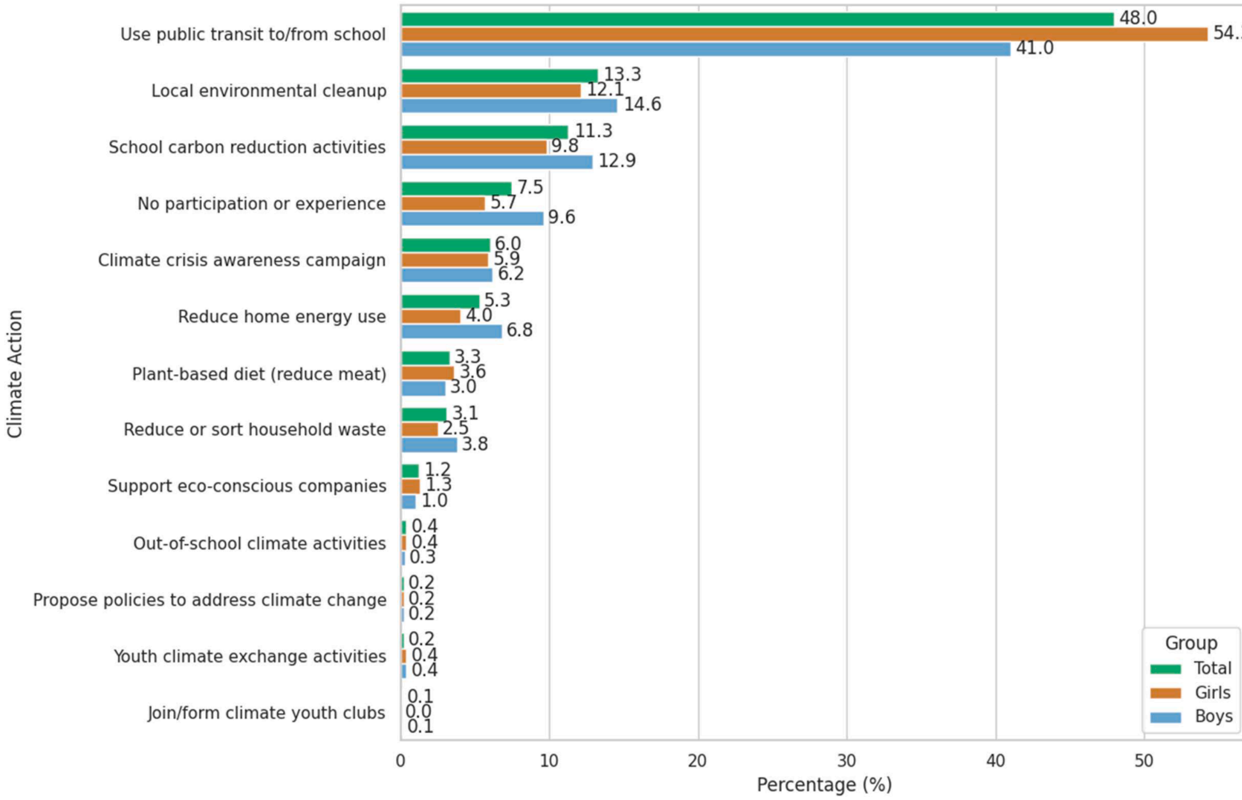


**Fig. 3.** Youth engagement in climate action by gender.

tailored climate education and engagement strategies by gender that acknowledge these distinct emotional and behavioral orientations.

We found gender differences in climate belief, with girls being more likely to believe that climate change is currently happening compared to boys. This aligns with a growing body of research showing that girls and women tend to express greater concern and awareness about climate-related risks than their male counterparts [18,23]. However, the majority of youth in our sample believed that it is currently happening (83 %) and, regardless of gender, most youth (95 %) attributed climate change to human activity like burning fossil fuels, suggesting that climate skepticism or denial is minimal in this population. This finding is encouraging and may reflect the success of global and national educational efforts and media discourse in shaping climate literacy among younger generations. The overall high level of belief in anthropogenic climate change among youth also highlights a generational shift in attitudes, which could have significant implications for future civic engagement and climate action. Nevertheless, the persistence of gender differences in belief suggests that sociocultural factors, including differential exposure to climate education, socialization patterns, and gendered emotional expression, may still play a role in shaping how youth internalize and respond to the climate crisis by gender.

A notable gender gap was observed in emotional and behavioral responses to climate change. Previous research consistently indicated that girls are more likely to engage emotionally with environmental concerns than boys, often reporting higher levels of worry, sadness, and distress in response to climate-related risks and future uncertainties [24–26]. These findings are consistent with the well-established literature on gender differences in emotional stability, neuroticism, and mental health, which suggests that women tend to report greater anxiety and emotional sensitivity across contexts [27]. Our results extend this general pattern to climate-related matters, illustrating how these gendered psychological tendencies intersect with societal norms around emotional expression and with the disproportionate environmental and social burdens women often face, particularly those in vulnerable communities such as urban settings and low-income households [28,29]. Among youth, it is also reported that girls may face additional barriers, such as gendered expectations to help with household chores during heatwaves and restrictions on outdoor play and mobility [30,31]. These findings may suggest that gendered climate vulnerabilities are not biologically predetermined but are instead produced and reinforced through broader systems of inequality. In addition to social positions, previous studies indicated that individuals who care more deeply about environmental issues or who have direct or lived experience with climate impacts [32], such as witnessing environmental degradation, displacement, or resource scarcity, are more likely to experience heightened levels of climate anxiety [33]. This includes anxiety related to the anticipated future, feelings of helplessness or grief, and moral distress about perceived inaction [33]. It is important to note that these emotional responses are not inherently negative; in fact, they can serve as important motivators for engagement and advocacy if acknowledged and supported [34,35].

Another highlight of this work that builds on recent literature is the recognition that physical activity promotion can serve as a powerful and accessible strategy to support youth participation in climate action, with broad co-benefits for both individual health and environmental sustainability [36–38]. Notably, active modes of transportation (e.g., walking or biking) offer a concrete and inclusive way for youth to contribute to climate mitigation while simultaneously improving varying health outcomes [39]. Furthermore, the 2025 Position Statement on Active Outdoor Play highlights the important role of active outdoor play in fostering environmental stewardship and supporting climate change adaptation and resilience [38,40,41]. In this study, using public transit, which often involves active transportation components (e.g., walking to bus stops or subway stations), was identified as the most common climate action taken by Korean youth (48 %). This was followed by local environmental cleanup activities (13 %), which also involve physical activity. These findings suggest that youth are already engaging in behaviors that are both climate-friendly and health-promoting, even if not

**Table 3**
Gender differences in the psychosocial and behavioral profiles of climate change.

| | Girls (Reference group) vs boys Odds Ratio (95 % Confidence interval) |
|---|---|
| **Cognitive Factors** | |
| *Climate belief* | |
| Currently happening | 0.68 (0.48–0.97)* |
| *Causal attribution* | |
| Human activity vs natural phenomenon | 1.17 (0.66–2.07) |
| *Climate Knowledge (know)* | |
| Net-Zero | 1.47 (1.17–1.85)** |
| Paris agreement | 0.94 (0.66–1.34) |
| Greenhouse gas emission reduction | 1.13 (0.90–1.41) |
| Sustainable development | 1.01 (0.77–1.32) |
| Global warming | 0.96 (0.71–1.31) |
| Climate refugee | 1.02 (0.84–1.24) |
| Climate inequality | 0.96 (0.77–1.20) |
| **Experiential Factors** | |
| *Climate awareness* (yes) | 0.93 (0.64–1.36) |
| *Climate anxiety* (high) | 0.63 (0.41–0.95)* |
| *Environmental Competency* (high) | 0.74 (0.53–1.05) |
| **Sociocultural Factors** | |
| *Sufficiency of school-based climate education* (yes) | 1.16 (0.95–1.43) |
| *National climate response effectiveness* (yes) | 2.19 (1.67–2.88)*** |
| *Optimism about solving climate change* (yes) | 1.96 (1.36–2.71)** |
| *Prioritized climate solution* | |
| Individual level info | 1.05 (0.79–1.40) |
| Climate change education | 1.07 (0.80–1.44) |
| Youth engagement | 0.98 (0.65–1.47) |
| Stronger penalties | 1.01 (0.65–1.58) |
| Incentives for action | 0.92 (0.64–1.32) |
| Green business practices | 1.00 (0.75–1.34) |
| Global cooperation | 1.01 (0.70–1.45) |
| Government policies | 0.95 (0.67–1.35) |
| **Behavioral factors** | |
| *Behavioral modification* (yes) | 0.74 (0.57–0.94)* |
| *Climate action* | |
| Activism[a] | 1.33 (0.95–1.88) |
| Pro-environmental behavior[b] | 0.64 (0.46–0.88)* |
| No participation | 1.82 (1.14–2.91)* |

[a] Activism included participating in school carbon reduction activities, conducting climate crisis awareness campaign, participating in youth climate exchange activities, proposing policies to address climate change, joining or forming climate youth clubs, and participating in out-of-school climate activities.

[b] Pro-environmental behavior included using public transit to/from school, reducing meat consumption (i.e., plant-based diet), engaging in local environmental cleanup, supporting eco-conscious companies, reducing home energy use, or reducing or sorting household waste.

Statistical significance:.
* $p < .05$.
** $p < .01$.
*** $p < .001$.

explicitly framed as such. This opens up an opportunity to design co-benefit interventions that promote physical activity for health while fostering climate literacy and environmental stewardship. Importantly, such approaches are inherently inclusive, allowing youth of all genders and backgrounds to engage meaningfully in climate action through everyday practices. Given the psychosocial and mental health benefits of physical activity such as enhanced self-efficacy, reduced anxiety, and improved mood [42], integrating physical activity-based strategies, particularly active outdoor play and active transportation, into climate engagement efforts may empower and enable youth to take on their preferred, accessible, health-promoting actions.

The findings of this work have several practical implications. Understanding gendered differences in perception and engagement with climate change can inform the design of targeted interventions and programs for youth. For instance, educators, climate communicators, and program developers can use this information to tailor curricular and activities that are sensitive to these differences to foster engagement and resilience among young people. Furthermore, policymakers and youth support organizations can also benefit from our study results by developing policies and initiatives that address emotional health while promoting adaptive behaviors in response to climate change. Relevant to our findings, a recent study among Korean adolescents aged 14–19 years suggested that cultivating self-efficacy and constructive attitudes toward climate change may empower youth to engage in climate governance [15]. Clarifying the mechanisms that motivate youth to engage in climate action, such as the roles of awareness, efficacy, attitudes, and concerns, is warranted in future studies.

This study draws on a representative sample of Korean youth, enabling a nuanced analysis of youth perspectives on climate change. By incorporating a gender-based analysis, the study adds to the growing body of literature that examines how identity and social norms shape young people's climate beliefs, emotional responses, and behaviors. The breadth of climate dimensions explored, from general awareness topics such as global warming to more complex and justice-oriented issues like Net-Zero, climate inequality, and the Paris Agreement, provides a comprehensive view of youth climate literacy. Nevertheless, the reliance on self-reported data introduces the possibility of social desirability bias or inaccuracies in how respondents perceive and report their behaviors and emotions. While the study explored gender differences and general demographic patterns, it did not account for factors such as household and parental factors (e.g., household income, parental education) or personal experiences with climate-related events, as these data were not available in the dataset used. Future work should measure and incorporate these potentially important factors. Youth are more likely to relate to and act on climate change when it is presented within the context of their everyday environments, educational experiences, and national discourse. Therefore, grounding climate education and policy communication in locally meaningful examples may enhance resonance, agency, and sustained engagement among young people.

## 5. Conclusions

Our study highlights that psychosocial and behavioral responses to climate change are embedded in and should be interpreted within the broader context of the social climate surrounding climate change, an evolving collective social environment largely shaped by media, politics, education, and interpersonal dynamics. Addressing these social influences through tailored messaging, education, and participatory programming could help close gender gaps in climate engagement and ensure that all youth, regardless of gender, are equally supported and empowered to act on the climate crisis. Encouraging boys to translate their trust for government and climate optimism into action, while reinforcing girls' agency and addressing the emotional labor often associated with their heightened sense of responsibility, may help promote more equitable and collective participation in climate solutions.

## Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

## CRediT authorship contribution statement

**Eun-Young Lee:** Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Leigh M. Vanderloo:** Writing – review & editing, Validation, Methodology, Conceptualization. **Guy Faulkner:** Writing – review & editing, Validation, Methodology, Conceptualization. **Justin Y. Jeon:** Writing – review & editing, Validation, Methodology, Conceptualization. **Ryan E. Rhodes:** Writing – review & editing, Validation, Methodology, Conceptualization. **John C. Spence:** Writing – review & editing, Validation, Methodology, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgement

The authors thank the National Youth Policy Institute (NYPI) for providing access to the data used in this study. The data are available for authorized use and can be found at h ttps://w ww.nypi.re.kr/archive/mps. The copyright and intellectual property rights of the data belong exclusively to NYPI and are protected under South Korean copyright law and international copyright treaties.

## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.joclim.2026.100663.

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